ACP-Gini: An Ancestor-Correlation Penalized Splitting Criterion for Multicollinearity-Robust Decision Trees

Huynh Khang Lam | Tuan Dat Phan | Trong Nhan Nguyen
FPT University, Vietnam

Abstract. Decision trees avoid coefficient-identification problems associated with linear regression, but correlated predictors can still make their selected feature sets redundant and their importance profiles difficult to interpret. We introduce ACP-Gini, a single-tree splitting criterion that multiplies each candidate's raw Gini gain by a product of correlation penalties against distinct features already used on the root-to-node path. The implementation reduces exactly to CART at alpha=0, excludes self-penalties, and uses raw gain for stopping. Evaluation combines a controlled correlated-feature generator with five real binary-classification data sets, 15 repeated cross-validation folds, and 50 bootstrap refits per fold. At selected alpha, ACP-Gini reduces bootstrap weighted redundancy by 12.1% on WDBC, 29.6% on Wine, 12.9% on Ionosphere, and 10.8% on Sonar, while predictive differences from CART are not statistically significant. A real-data operating sweep shows that WDBC at alpha=0.6 reduces weighted redundancy by 38.5% over three seeds with a 0.17 percentage-point accuracy change; the seed-42 five-fold result is 46.5% with identical accuracy. Synthetic group coverage and within-group concentration increase from alpha=0 to 1 at every tested correlation level, with a small rise in noise importance. Stability improves on Wine and Pima but decreases on WDBC and Sonar, revealing a stability--redundancy tradeoff rather than a uniform stability gain.

Keywords: Decision trees ; multicollinearity ; feature redundancy ; model stability ; interpretable machine learning

Introduction

Decision trees are often described as ``immune'' to multicollinearity because a split uses one predictor at a time. This claim is narrower than it first appears. Trees do avoid simultaneous coefficient estimation, yet greedy induction must still choose among correlated variables with nearly interchangeable gains. Small sample changes can alter the representative chosen from a correlated block, while a single fitted tree may repeatedly use several members of the same block along different branches. Consequently, prediction can remain strong while the selected feature set is internally redundant and impurity-based importance is spread across substitutes.

The distinction matters for interpretable machine learning. A model can be accurate but give a needlessly repetitive account of the evidence used for its decisions. Correlated-feature effects on variable importance are well documented for forests [4, 5, 7], while tree instability motivated resampling methods such as bagging [9]. For an auditable single tree, however, preprocessing by variance-inflation-factor (VIF) pruning discards variables globally, and regularized-tree methods apply feature-level penalties that do not express the pairwise relationship between a candidate and the path that leads to the current node.

ACP-Gini addresses this local redundancy problem. At a node T, it discounts the Gini gain of candidate X_j according to its absolute correlation with distinct features used by ancestors of T. The design retains the familiar CART induction procedure [1, 11], remains exactly backward-compatible at alpha=0, and leaves raw gain responsible for deciding whether an informative split exists.

This study makes four contributions:

- an ancestor-correlation penalty for a single interpretable tree, including explicit no-self-penalty and raw-gain stopping rules;
- two direct measures of selected-set redundancy, plus bootstrap stability and split-composition measures;
- controlled group-level synthetic evaluation and a five-data-set real benchmark against CART, VIF+CART, and an RRF-inspired single tree;
- a reproducible from-scratch implementation whose tables and figures are generated from committed CSV outputs.

The scope is interpretable model construction and model diagnostics, complementing the broader argument for directly usable interpretable models in consequential settings [12].

Related Work

CART and instability.

CART greedily maximizes an impurity decrease at each node [1]. Greedy induction is sensitive to perturbations because an early alternative split changes all descendants. Li and Belford studied instability across decision-tree induction algorithms

[10]; bagging was motivated partly by exploiting the variance of unstable learners [9]. These results concern predictive variance and structure, but do not directly regulate correlation among selected split features.

Correlated predictors and importance.

Conditional permutation importance was proposed to address dependencies among predictors [4]. Nicodemus et al. demonstrated that correlation changes permutation-importance behavior [5]; Genuer et al. examined variable selection in random forests [6]; and Louppe et al. analyzed impurity-based importance in randomized trees [7]. These approaches are mainly diagnostic or ensemble-oriented. ACP-Gini changes the construction of one tree and retains unpenalized impurity gain for comparable feature-importance accounting.

Feature regularization and preprocessing.

Regularized Random Forest penalizes the gain of newly introduced features [2], and guided RRF incorporates external importance information [3]. Our RRF-style baseline is explicitly a single-tree adaptation so that all splitting methods share one implementation. VIF pruning represents the preprocessing family; the threshold of 10 follows common practice summarized by Dormann et al. [8]. Unlike global removal, ACP-Gini can retain a correlated predictor for a branch where it supplies useful conditional information.

The gap is therefore specific: prior work does not inject an interpretable pairwise correlation with root-to-node ancestors into the split criterion of a single tree. ACP-Gini is not an ensemble replacement; it is a local criterion for reducing repetitive feature use while preserving the mechanics and inspectability of CART.

Methodology

Let T contain n_T samples and let T_L,T_R be a candidate split. With class proportions p_k(T), Gini impurity and raw gain are

For each feature, the best threshold is found by sorting its values and evaluating midpoints between distinct adjacent values using cumulative class counts.

Let A(T) be the set of distinct feature indices used by ancestors on the root-to-T path. ACP-Gini excludes the candidate itself from the penalty set:

The selected candidate maximizes (S,Delta G,-j) lexicographically. Thus ties first favor larger raw gain and then the lowest feature index. Repeated splits on the same continuous predictor are not spuriously multiplied by (1-alpha).

Properties.

For fixed raw gain, the score is non-increasing in each absolute ancestor correlation and in alpha. The product accumulates moderate redundancy across several distinct ancestors. For alpha<1, all factors remain positive, so no feature is excluded by a hard threshold. At alpha=0, every factor equals one and the candidate ordering is exactly CART's ordering. Importantly, stopping depends only on whether any candidate has positive raw gain; the penalty reorders informative candidates but never turns an informative node into a leaf.

Pearson correlation is computed once on the full training fold by default. Spearman correlation rank-transforms columns before the same calculation. Constant-column NaNs are replaced by zero correlation. A node-local ablation recomputes correlations when at least 30 samples reach the node and otherwise falls back to the global matrix. Global preprocessing costs O(nd^2) once; lookup of ancestor correlations adds O(d,|A(T)|) per node beyond the threshold scan.

Parameter selection.

Inner stratified folds evaluate alpha in 0.2,0.4,0.6,0.8,1.0. Eligible values have macro-F1 within one standard deviation of the alpha=0 mean, and importance rank correlation selects among eligible values. The final data-set value is the lower quartile of fold-level selections, rounded down to the grid. This conservative aggregation biases disagreements toward a weaker, more CART-like penalty, analogous in spirit to a one-standard-error rule. The RRF-style lambda in 0.5,0.7,0.9 uses the upper quartile because larger lambda is the weaker penalty.

Experimental Setup

We use WDBC from scikit-learn [13], Wine Quality, Ionosphere, and Sonar from UCI [14], and Pima Diabetes from OpenML. All are numeric binary-classification tasks. Pairwise |r|>0.7 occurs most frequently in WDBC and Sonar; Pima serves as a low-collinearity control. Cached CSV files accompany the code.

Synthetic data contain five independent Gaussian signals with weights (1,.8,.6,.4,.2), three correlated copies per signal, and five noise features. A copy is rho Z+sqrt1-rho^2epsilon for rho in .7,.8,.9,.95,.99. Group coverage is the fraction of the five signal groups represented among the top-five importance features. Group concentration averages max_j in gI_j/sum_j in gI_j over all five groups; a zero-mass group contributes zero. Noise importance is the total mass on the five noise columns.

All main methods use depth 6 and minimum leaf size 5. Three shuffled five-fold runs (seeds 42--44) provide 15 paired observations. Each training fold is additionally bootstrapped 50 times. Predictive metrics are accuracy, macro-F1, and AUC. Stability metrics comprise Jaccard similarity of all used-feature sets, top-five-importance Jaccard, Spearman correlation of importance vectors, and split-composition distance, defined as one minus multiset Jaccard over (depth,feature) internal-node pairs.

For a fitted tree with used set U, selected-set redundancy is the mean |r_ij| over pairs in U. Importance-weighted redundancy is

Both use the full-data correlation matrix for a consistent diagnostic scale and are reported for main fits and averaged bootstrap fits. Wilcoxon signed-rank tests use paired fold observations [15]. Root agreement is retained in raw CSVs but omitted from headline tables because ACP-Gini is identical to CART at the root by construction.

Results and Discussion

Correctness and Synthetic Ground Truth

The alpha=0 sanity suite passed all 15 seed--data-set cases. Pearson and Spearman modes produced identical structures and predictions when the penalty was inactive; held-out accuracy differed from scikit-learn's tree by at most 0.01.

Figure~fig:synthetic reports group-level results. At rho=.9, group coverage increases from .68 to .74 and group concentration from .690 to .745 between alpha=0 and 1. Across the five rho values, endpoint coverage gains range from .04 to .10 and endpoint concentration gains from .027 to .098. Intermediate values are not strictly monotone for every rho, but every endpoint improves. Figure~fig:sensitivity shows that accuracy remains near .79 at rho=.9, while noise importance rises from .018 to approximately .033. This is a small but real side effect: discouraging redundant signal variables occasionally exposes weak noise splits.

Exact-original recovery is less suitable at high rho: an original and its copy are close substitutes, and selecting one representative should not be counted as failure merely because it is not the designated original column. The group metrics align the evaluation with the method's intended behavior.

Prediction and Selected-Set Redundancy

The conservative selected values are .2 (WDBC), 1.0 (Wine), .4 (Ionosphere), .2 (Sonar), and .2 (Pima). Accuracy differences from CART are statistically indistinguishable on every data set. Sonar's direction is negative but shrinks to only .65 percentage points after lower-quartile aggregation; Ionosphere changes from .885 to .886.

The primary result is lower internal correlation. Bootstrap weighted redundancy falls from .621 to .545 on WDBC (12.1%), .183 to .128 on Wine (29.6%), .168 to .146 on Ionosphere (12.9%), and .218 to .195 on Sonar (10.8%). All four reductions are significant at p<.001. Pima changes by only .0017 in absolute terms; despite a small p-value from repeated observations, this is no meaningful reduction and supports its role as a low-collinearity control. Figure~fig:redundancy visualizes the comparison.

The operating sweep in Fig.~fig:sweep separates a conservative default from attainable operating points. Across all three seeds, WDBC at alpha=.6 reduces main-fit weighted redundancy from .634 to .390 (38.5%) while accuracy changes from .934 to .932. The preregistered seed-42 five-fold check reproduces .656 to .351 (46.5%) with accuracy unchanged at .926. Wine shows a smooth redundancy decline from .166 to .118 while accuracy stays between .746 and .748. Thus practitioners can increase alpha when redundancy reduction is worth a small tolerance in predictive performance.

The Stability--Redundancy Tradeoff

Importance rank correlation rises by .179 on Wine and .017 on Pima; top-five Jaccard rises by .069 and .010, respectively. Wine's gains are significant at p=.0001. Conversely, rank stability decreases by .057 on WDBC and .040 on Sonar, with smaller negative changes on Ionosphere. Figure~fig:stability shows this conditional pattern.

This is not equivalent to predictive failure. CART's apparent stability can arise from repeatedly selecting the same correlated block, whereas ACP-Gini deliberately diversifies downstream evidence. Lower redundancy and higher cross-bootstrap rank agreement are distinct goals. ACP-Gini improves both where redundant choice drives instability (Wine), but trades rank consistency for less correlated feature use in block-correlated WDBC and Sonar.

Ablation, Runtime, and Qualitative Analysis

The ablation does not identify a universally dominant configuration. Spearman obtains the highest WDBC accuracy (.937), while Pearson-min is strongest on synthetic accuracy (.796); product-Pearson-global remains the simplest preregistered default. Global-scope runtime ranges from .85 to 1.33 times CART across data sets, and node-local scope from .84 to 1.44 times. The overhead is data dependent and is not uniformly below 10%.

Figure~fig:trees illustrates that the WDBC root and left branch remain identical, while the right child changes from mean texture to texture error under ACP-Gini. Average node counts remain comparable because raw-gain stopping preserves informative splitting: for example, WDBC uses 24.2 nodes for ACP-Gini and 24.7 for CART.

Across the WDBC repeated folds, CART and ACP-Gini disagree on 65 predictions. CART alone is correct in 33 cases and ACP-Gini alone in 32; 20 disagreements are near a [.4,.6] probability boundary. The near balance is consistent with unchanged aggregate accuracy.

Research questions.

RQ1 is conditional: stability improves on Wine and Pima but decreases on WDBC and Sonar. RQ2 is supported by synthetic group metrics and the lower selected-set redundancy on four collinear real data sets. RQ3 is supported: selected settings have statistically indistinguishable accuracy, while the sweep exposes the adjustable tradeoff. RQ4 is not supported as originally framed: node counts and depth are comparable; compactness is better characterized as lower redundancy of the selected evidence.

Limitations.

ACP-Gini is a greedy heuristic without a global optimality guarantee. Correlation does not distinguish redundancy from useful conditional interaction. High alpha can increase noise importance and, on WDBC at alpha=1, reduce accuracy to .921. Results concern numeric binary data and one tree; categorical association measures such as Cramer's's V and ensemble integration remain future work. The five data sets provide varied correlation structures but do not establish universal behavior.

Conclusion

ACP-Gini introduces an interpretable path-dependent correlation penalty while preserving CART behavior at alpha=0, repeated-feature splitting, and raw-gain stopping. The strongest evidence concerns redundancy rather than a uniform stability gain: selected feature sets are less internally correlated on four collinear real data sets without statistically significant predictive loss, and synthetic group-level recovery improves across all tested correlation endpoints. Stability depends on data structure, yielding an informative stability--redundancy tradeoff. Future work should study theoretical conditions for beneficial penalties, categorical association, regression trees, and ensemble variants.

Generated Evidence Tables

Table 1. Data-set statistics.

dataset	n	d	positive_rate	pair_fraction_abs_r_gt_07
WDBC	569	30	0.627	0.161
Wine	6497	11	0.633	0.018
Ionosphere	351	34	0.359	0.005
Sonar	208	60	0.534	0.031
Pima	768	8	0.349	0.000

Table 2. Mean predictive performance over 15 folds.

dataset	method	accuracy	macro_f1	auc
Ionosphere	ACP-Gini	0.886	0.876	0.904
Ionosphere	CART	0.885	0.874	0.910
Ionosphere	RRF-style	0.885	0.875	0.910
Ionosphere	VIF+CART	0.885	0.874	0.910

Pima	ACP-Gini	0.733	0.692	0.774
Pima	CART	0.737	0.698	0.774
Pima	RRF-style	0.735	0.696	0.776
Pima	VIF+CART	0.737	0.698	0.774
Sonar	ACP-Gini	0.731	0.729	0.764
Sonar	CART	0.737	0.736	0.752
Sonar	RRF-style	0.726	0.724	0.750
Sonar	VIF+CART	0.715	0.712	0.754
WDBC	ACP-Gini	0.934	0.929	0.957
WDBC	CART	0.934	0.929	0.953
WDBC	RRF-style	0.941	0.936	0.958
WDBC	VIF+CART	0.931	0.926	0.962
Wine	ACP-Gini	0.746	0.726	0.800
Wine	CART	0.747	0.725	0.797
Wine	RRF-style	0.748	0.724	0.799
Wine	VIF+CART	0.747	0.724	0.798

Table 3. Bootstrap selected-set redundancy.

dataset	method	bootstrap_set_redundancy	bootstrap_weighted_redundancy
Ionosphere	ACP-Gini	0.159	0.146
Ionosphere	CART	0.183	0.168
Ionosphere	RRF-style	0.183	0.166
Ionosphere	VIF+CART	0.182	0.167
Pima	ACP-Gini	0.169	0.178
Pima	CART	0.170	0.180
Pima	RRF-style	0.169	0.180
Pima	VIF+CART	0.170	0.180
Sonar	ACP-Gini	0.204	0.195
Sonar	CART	0.234	0.218
Sonar	RRF-style	0.226	0.216
Sonar	VIF+CART	0.220	0.195
WDBC	ACP-Gini	0.338	0.545
WDBC	CART	0.465	0.621
WDBC	RRF-style	0.458	0.603
WDBC	VIF+CART	0.283	0.343
Wine	ACP-Gini	0.220	0.128
Wine	CART	0.245	0.183
Wine	RRF-style	0.244	0.181
Wine	VIF+CART	0.238	0.169

Table 4. Bootstrap stability metrics.

dataset	method	feature_set_jaccard	top5_jaccard	importance_rank_corr	structural_distance
Ionosphere	ACP-Gini	0.322	0.293	0.372	0.880
Ionosphere	CART	0.335	0.303	0.388	0.878
Ionosphere	RRF-style	0.319	0.304	0.378	0.882
Ionosphere	VIF+CART	0.338	0.305	0.393	0.877
Pima	ACP-Gini	0.899	0.677	0.714	0.663
Pima	CART	0.928	0.668	0.698	0.664
Pima	RRF-style	0.862	0.671	0.701	0.669
Pima	VIF+CART	0.928	0.668	0.698	0.664
Sonar	ACP-Gini	0.122	0.106	0.088	0.953
Sonar	CART	0.149	0.111	0.128	0.947
Sonar	RRF-style	0.128	0.111	0.112	0.952
Sonar	VIF+CART	0.166	0.128	0.155	0.945
WDBC	ACP-Gini	0.268	0.224	0.241	0.908
WDBC	CART	0.306	0.250	0.298	0.899
WDBC	RRF-style	0.246	0.239	0.265	0.901
WDBC	VIF+CART	0.311	0.266	0.308	0.884
Wine	ACP-Gini	0.898	0.675	0.845	0.512
Wine	CART	0.986	0.606	0.666	0.568
Wine	RRF-style	0.953	0.603	0.666	0.573
Wine	VIF+CART	0.993	0.620	0.766	0.549

Figures

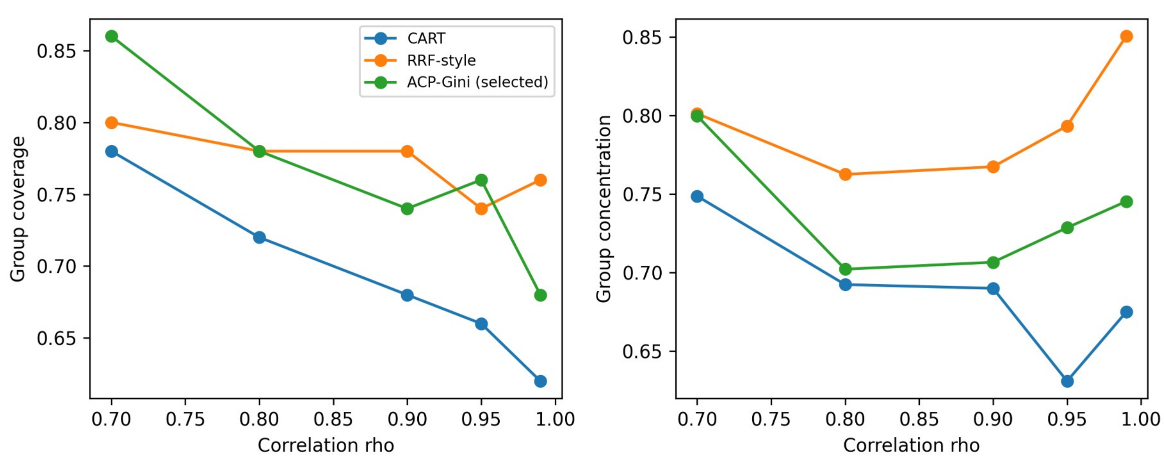

Figure 2. Synthetic group coverage and concentration.

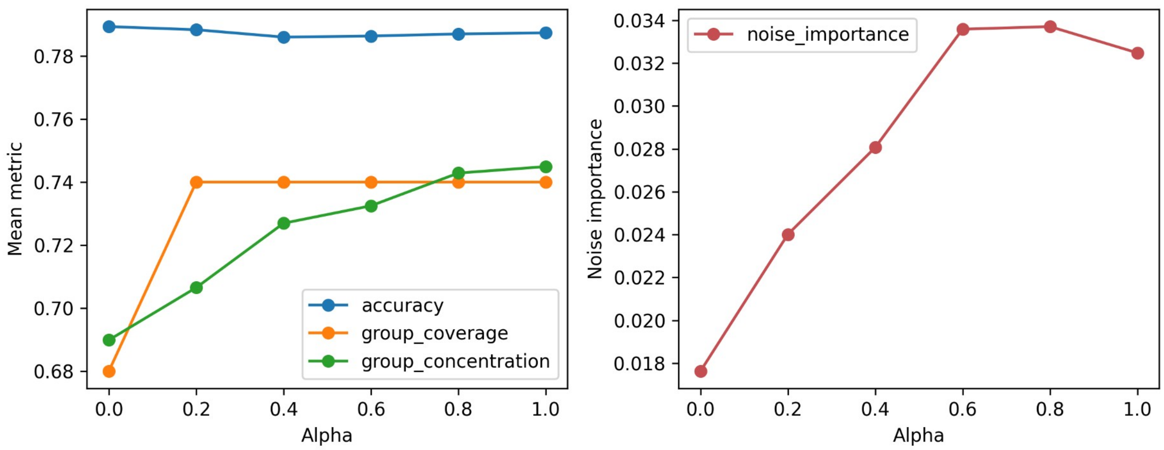

Figure 3. Alpha sensitivity and noise importance.

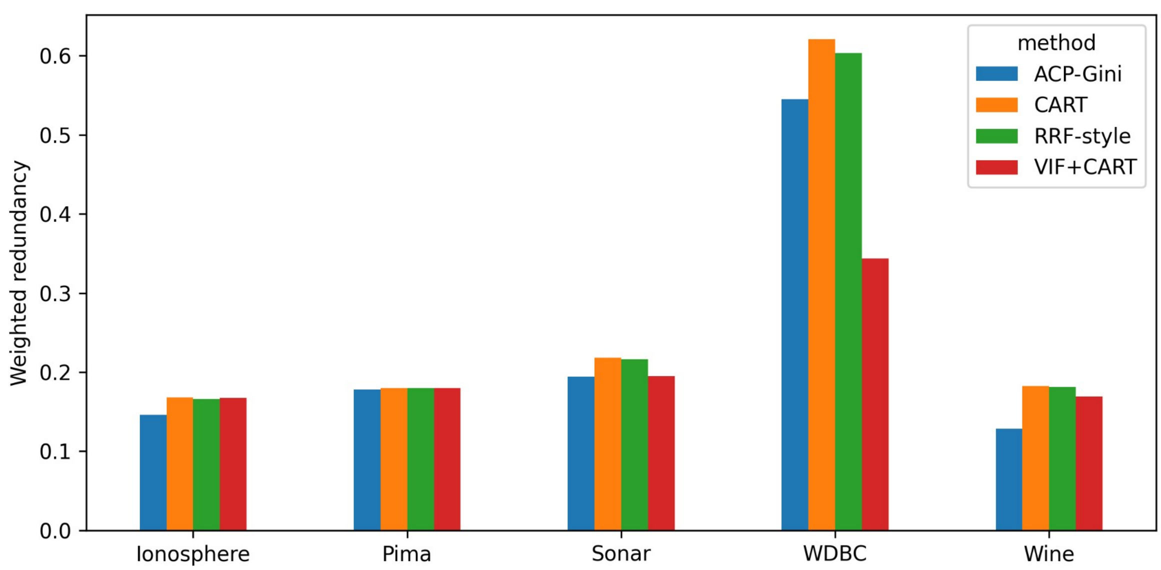

Figure 4. Bootstrap weighted redundancy.

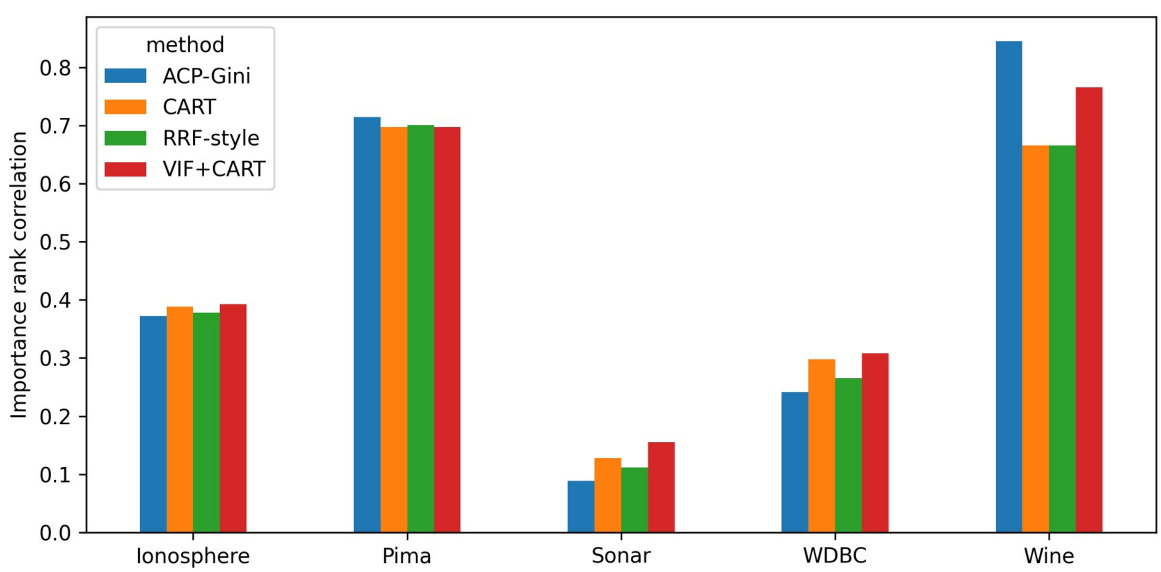

Figure 4b. Conditional importance stability.

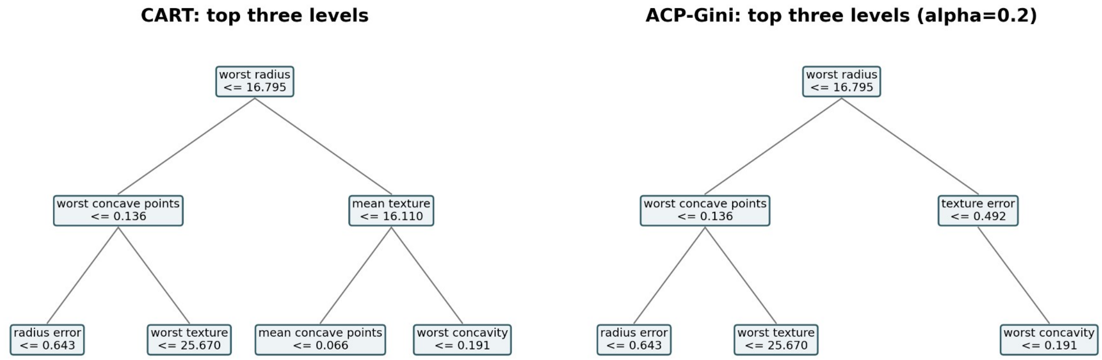

Figure 5. WDBC tree comparison.

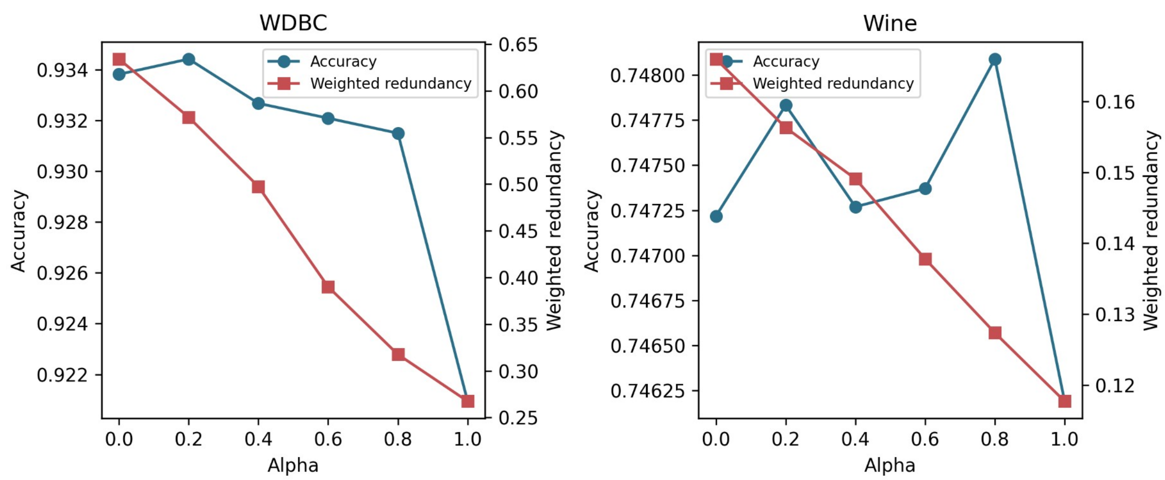

Figure 6. WDBC and Wine operating characteristics.

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